

**DEEPPFAKE IMAGE DETECTION USING
COLOR CHANNEL CO-OCCURRENCE
MATRICES AND CONVOLUTIONAL
NEURAL NETWORKS**



**MINI PROJECT REPORT
(23EXC302)**

Submitted by

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COIMBATORE**

ANNA UNIVERSITY, CHENNAI

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BONAFIDE CERTIFICATE

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SCHOOL OF COMPUTER APPLICATIONS

VISION

To be a nationally prominent and recognized department in academics and research activities with the aim of developing competent professionals to serve the society in the ever-changing industry.

MISSION

- Enabling the students to be knowledgeable and creative by developing state of the art curriculum and innovative teaching methods.
- Producing highly competent professionals by providing training programs to bridge the gap between the institution and industry.
- Enhancing the quality of research and productivity by providing the required facilities.

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- **PEO 1:** Graduates will be competent enough to identify, analyse, design and develop software solutions to the problems in the field.
- **PEO 2:** Graduates, equipped with good communication skills and technical expertise will be able to conceptualize and lead projects with an aim to become effective entrepreneurs or employees.
- **PEO 3:** Graduates, with exposure to highly active research environment, will have the motivation to participate and contribute in research and development.

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STUDENTS IN THE MCA PROGRAMME WILL HAVE THE ABILITY TO

1. Apply knowledge of computing fundamentals, computing specialization, mathematics, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.
2. Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
3. Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
4. Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.
6. Understand and commit to professional ethics and cyber regulations, responsibilities, and norms of professional computing practices.
7. Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.

8. Demonstrate knowledge and understanding of the computing and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
9. Communicate effectively with the computing community, and with society at large, about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.
10. Understand and assess societal, environmental, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practices.
11. Function effectively as an individual and as a member or leader in diverse teams and in multidisciplinary environments.
12. Identify a timely opportunity and using innovation to pursue that opportunity to create value and wealth for the betterment of the individual and society at large.

DECLARATION



I hereby declare that this Project report entitled “**DEEPFAKE IMAGE DETECTION USING COLOR CHANNEL CO-OCCURENCE MATRICES AND CONVOLUTIONAL NEURAL NETWORKS**” submitted by me for the degree of **Master Of Computer Applications at Karpagam College of Engineering, Coimbatore** is the record of original work done by me under the guidance and supervision of **Dr. P. SHANTHI, Professor** at the School of Computer Applications, Karpagam College of Engineering, Coimbatore –641032 and has not formed the basis for the award of any degree, or diploma or titles in this Institution or any other Institution of higher learning.

(Signature of the Candidate)

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Date:

Place: Coimbatore

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VELMURUGAN U

ABSTRACT

The rapid advancement of deep learning and Generative Adversarial Networks (GANs) has led to the emergence of highly realistic manipulated images and videos known as deepfakes. These synthetic media pose serious ethical, security, and social threats by enabling the creation of deceptive content that is nearly indistinguishable from authentic visuals. As a result, the detection of deepfakes has become a critical research domain in computer vision and multimedia forensics. This study presents a comprehensive comparative analysis of existing deepfake image detection models developed between 2018 and 2025, emphasizing their methodologies, performance, and limitations. The comparison encompasses three major categories of detection techniques: handcrafted feature-based models such as Local Binary Pattern (LBP) and Co-occurrence Matrix (GLCM), deep learning-based architectures including MesoNet, XceptionNet, and EfficientNet, and hybrid frameworks that integrate both statistical and convolutional features. Each method is evaluated based on its feature extraction approach, dataset dependency, computational complexity, and overall accuracy. Findings from the reviewed literature reveal that handcrafted models provide interpretability and efficiency but lack scalability, while deep learning systems achieve high accuracy at the cost of computational resources. The study concludes that hybrid frameworks—particularly those combining Co-occurrence Matrix and Convolutional Neural Networks (CNN) offer the most balanced trade-off between performance, generalization, and interpretability. These models consistently achieve accuracies exceeding 89%, outperforming traditional handcrafted approaches and demonstrating robustness under varying compression and lighting conditions. This comparative study serves as a consolidated reference for future research, providing key insights into the evolution, effectiveness, and challenges of deepfake image detection methodologies.

Keywords — Deepfake detection, feature extraction, image forensics, convolutional neural networks, co-occurrence matrix, transformer models, texture analysis, frequency-domain features, hybrid detection methods, generative adversarial networks, comparative analysis, image manipulation, digital forensics, detection accuracy, deep learning approaches, dataset analysis.

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ABBREVIATIONS

- AI – Artificial Intelligence
- API – Application Programming Interface
- AUC – Area Under Curve
- CNN – Convolutional Neural Network
- CPU – Central Processing Unit
- DL – Deep Learning
- DF – Deepfake
- DFDC – DeepFake Detection Challenge
- DFT – Discrete Fourier Transform
- DNN – Deep Neural Network
- FF++ – FaceForensics++
- GAN – Generative Adversarial Network
- GLCM – Gray Level Co-occurrence Matrix
- GPU – Graphics Processing Unit
- LBP – Local Binary Pattern
- ML – Machine Learning
- MLP – Multi-Layer Perceptron
- PCA – Principal Component Analysis
- PSNR – Peak Signal-to-Noise Ratio
- ReLU – Rectified Linear Unit
- ROC – Receiver Operating Characteristic
- ROC-AUC – Receiver Operating Characteristic - Area Under Curve
- SVM – Support Vector Machine

CHAPTER – 1

INTRODUCTION

Deepfake image synthesis has rapidly evolved over recent years due to advances in generative adversarial networks, autoencoders, diffusion models, and face-swapping algorithms. These technologies enable the creation of highly realistic manipulated facial images that are difficult to distinguish from authentic photographs using the naked eye. The increasing accessibility of deepfake tools has raised concerns in digital forensics, media verification, cybersecurity, and social communication. In response, researchers have proposed a wide spectrum of detection techniques that analyze visual inconsistencies left by generative models. These techniques range from handcrafted texture descriptors to sophisticated deep learning architectures, each contributing distinct perspectives toward understanding and detecting visual manipulation. The body of existing literature reveals noticeable diversity in both feature extraction strategies and model design philosophies. Some studies rely on classical statistical descriptors such as gray-level co-occurrence matrices, local binary patterns, and color–texture correlations, aiming to capture irregular pixel relationships that arise during facial synthesis. Other works adopt deep convolutional architectures that learn hierarchical features directly from images, enabling the detection of subtle structural deviations and semantic inconsistencies. More recent studies introduce transformer-based architectures and multimodal fusion mechanisms that expand the detection focus beyond local textures to global contextual information. As each technique operates under different assumptions and computational frameworks, their strengths and limitations vary considerably depending on the dataset, deepfake generation method, and experimental conditions.

A comparative perspective is therefore essential to understand the landscape of deepfake image detection. Rather than evaluating an individual model in isolation, this survey critically analyzes multiple published research papers to highlight how detection accuracy, robustness, and generalizability differ across approaches. The comparison spans feature extraction strategies, model architectures, datasets, training pipelines, evaluation metrics, and reported findings. By examining these variations side-by-side, the survey identifies patterns in performance, recurring limitations, and research gaps that influence the reliability of detection algorithms in real-world scenarios. This approach not only summarizes what each study contributes but also provides a broader understanding of how detection methods evolve and where improvements are still needed.

Ultimately, this survey aims to offer a consolidated view of the deepfake detection field by synthesizing insights from past work and presenting them in a structured comparative format. The goal is to support researchers, students, security practitioners, and forensic analysts in understanding which methods perform best under different circumstances, why certain techniques fail, and how future research can address these shortcomings. By emphasizing comparative analysis, the survey highlights key trends that shape the development of next-generation deepfake detection systems.

1.1 OBJECTIVES

The primary objective of this survey is to conduct a detailed comparative analysis of the different deepfake image detection techniques presented across multiple existing research papers. Instead of focusing on the development or implementation of a new model, this work aims to understand how existing methods differ in their assumptions, processing strategies, and overall effectiveness. The objective is to clearly outline the evolution of detection methods, showing how the field has progressed from traditional handcrafted features to modern deep neural networks and attention-based architectures. This comparison helps reveal why certain approaches perform better and under what circumstances their performance begins to decline. Another important objective is to examine the feature extraction strategies used across the selected studies. By comparing classical texture-based descriptors, frequency-domain patterns, CNN-derived features, and transformer-based representations, the survey aims to determine which types of features offer the strongest detection performance. This understanding is crucial, because deepfake manipulation introduces artifacts at different levels, and each feature category captures different types of inconsistencies. Highlighting these distinctions allows the survey to identify which techniques are most effective against modern GAN-generated content. A further objective is to evaluate the datasets, metrics, and evaluation procedures used by different papers. Many detection models achieve high accuracy only on specific datasets but fail to generalize across unseen deepfake styles. By comparing dataset diversity and testing conditions, the survey identifies gaps in generalizability and highlights the need for standardized evaluation protocols. This helps readers understand the difference between high accuracy achieved under controlled conditions versus performance in real-world settings. Finally, the survey aims to highlight limitations, open challenges, and future opportunities that emerge from the comparative discussion. By analyzing recurring issues—such as compression

sensitivity, poor robustness to unseen manipulation techniques, and limited dataset variety—the survey outlines clear directions for future research. These objectives collectively contribute to a comprehensive understanding of the deepfake detection landscape and serve as a foundation for future model development, dataset creation, and forensic applications.

1.2 PROJECT SCOPE

The scope of this survey extends across a wide range of deepfake image detection techniques that have emerged over the past decade, covering both classical approaches and modern deep learning models. This work focuses exclusively on image-based deepfake detection, analyzing how different research papers utilize feature extraction methods, network architectures, and evaluation strategies to identify tampered visual content. The survey includes studies that employ texture descriptors, statistical feature matrices, handcrafted visual cues, convolutional neural networks, hybrid deep learning models, and transformer-based frameworks. By examining papers that use varied datasets, experimental pipelines, and accuracy metrics, the survey aims to present a broad and unbiased comparison of existing methods. The purpose of this scope is to highlight how different feature extraction strategies influence detection performance. Some of the reviewed papers concentrate on statistical pixel-level relationships, while others extract high-level semantic features through deep neural networks. This survey explores the strengths, limitations, and distinctive characteristics of each approach, identifying how they respond to different types of manipulations and which methods perform consistently across diverse datasets. The comparative nature of this study ensures that readers gain insight into the relative performance of various models rather than the isolated success of individual techniques. The scope also includes the identification of research gaps and emerging challenges in the field of deepfake detection. Many models perform well in controlled environments but fail when tested on unseen manipulation techniques or compressed images. By compiling the limitations reported across multiple studies, the survey highlights common issues such as dataset bias, lack of generalization, and sensitivity to resolution changes. This helps define the practical boundaries of current detection approaches and guides future researchers toward potential improvements. Overall, this survey covers the technical landscape, performance variations, methodological differences, and forensic implications found in existing deepfake detection research. It does not propose or implement a new algorithm but instead focuses on understanding and evaluating the body of existing work through structured comparison.

The scope is designed to support students, researchers, and professionals who seek a clear and comprehensive understanding of how deepfake detection methods have evolved and how they differ in their effectiveness across real-world scenarios. The scope of this survey extends across a comprehensive examination of deepfake image detection techniques, focusing on how various research papers have approached the challenge of identifying synthetic facial images generated through modern AI-based manipulation. This study evaluates a wide range of methodologies published in recent years, highlighting how the field has evolved from early handcrafted features to sophisticated deep neural architectures.

CHAPTER – 2

EXISTING SYSTEM & SYSTEM ANALYSIS

2.1 EXISTING APPROACHES

In the early stages of deepfake detection research, models heavily relied on handcrafted feature extraction techniques that used statistical and texture-based descriptors to differentiate between real and manipulated images. Techniques such as the Gray-Level Co-occurrence Matrix (GLCM), Local Binary Pattern (LBP), and Principal Component Analysis (PCA) were widely used to capture the structural or textural inconsistencies introduced by deepfake synthesis [1][4][8]. For instance, the GLCM technique analyzes how often pixel intensity values occur in specific spatial relationships, extracting statistical measures such as contrast, energy, correlation, and homogeneity. These features effectively expose pixel-level irregularities that GANs often fail to reproduce naturally. Researchers such as Nataraj et al. [1] demonstrated that GLCM-based models could identify fake images with nearly 89% accuracy by focusing on texture variations, especially in low-frequency image regions. Similarly, Yang et al. [8] proposed a color-texture analysis framework that enhanced the detection of manipulated biometric images by measuring co-occurring pixel relationships across RGB channels. These approaches, though interpretable and computationally light, lacked adaptability to newer, high-quality deepfakes generated by advanced GAN architectures like StyleGAN and ProGAN. As deepfake content became more complex, the research community shifted toward Convolutional Neural Network (CNN)-based architectures that could automatically learn discriminative features directly from image data. Afchar et al. [2] introduced MesoNet, a compact CNN designed for facial forgery detection, which achieved strong performance with fewer parameters, making it ideal for smaller datasets. Later models such as XceptionNet and EfficientNet extended this approach by incorporating deeper convolutional layers, separable convolutions, and larger training datasets. These networks learned both global and local image representations, making them more effective against high-quality manipulations.

Recent studies such as those by Zhou et al. [7] and Gupta et al. [9] further expanded detection capabilities using frequency-domain CNNs and self-attention mechanisms. Frequency-domain approaches analyze the spectral inconsistencies introduced by GAN-generated images, as fakes often exhibit unnatural periodic artifacts. Transformer-based architectures, on the other hand, capture long-range dependencies within the image through attention mechanisms, improving

contextual understanding of forgeries. Despite their high accuracy—often exceeding 91%—these models require significant computational power, large-scale datasets, and extended training times. To address the limitations of both handcrafted and deep models, researchers proposed hybrid systems that combine the interpretability of texture-based features with the learning power of CNNs. Banerjee and Yadav [6] developed a CNN integrated with handcrafted features to achieve better generalization on unseen datasets. Similarly, hybrid frameworks leveraging GLCM and CNNs demonstrated stronger resilience to noise, compression, and illumination changes compared to single-model approaches. These findings highlight the value of integrating statistical analysis with deep feature extraction to improve model robustness and accuracy.

2.2 LIMITATIONS OF EXISTING APPROACHES

Despite the considerable progress made in deepfake image detection, existing approaches face several limitations that restrict their generalizability and long-term reliability. Many classical handcrafted methods struggle to adapt to modern GAN architectures, which produce visually consistent outputs with fewer detectable pixel-level artifacts. These techniques often perform well only on specific datasets and fail when evaluated against newer or unseen forgery methods. Deep learning-based models, although more powerful, also encounter challenges such as high computational complexity, overfitting, and sensitivity to compression or noise. A common limitation observed across multiple studies is dataset dependency; several models achieve high accuracy only within controlled environments but exhibit a sharp decline when tested on cross-dataset or real-world images. Additionally, many papers use limited training samples or imbalanced datasets, leading to biased performance assessments. Another recurring issue is the rapid evolution of deepfake generation techniques, which often outpace detection strategies, causing models to become obsolete quickly. These limitations collectively emphasize the need for more generalized, adaptive, and robust detection techniques capable of handling diverse manipulation styles and real-world distortions.

2.3 SYSTEM ANALYSIS

A systematic analysis of the reviewed techniques reveals clear distinctions in how researchers design and evaluate deepfake detection systems. Studies employing handcrafted features emphasize local pixel behavior, texture anomalies, and color inconsistencies, offering computational efficiency but limited adaptability. In contrast, CNN-based models focus on

learning hierarchical spatial representations, producing stronger detection performance, especially when high-quality training data is available. Transformer-based and hybrid models introduce long-range attention mechanisms and cross-modal analysis, expanding detection capabilities beyond localized features. The reviewed papers also differ significantly in dataset usage, with some relying on commonly available datasets such as FaceForensics++, while others introduce custom datasets with varying degrees of complexity. Evaluation procedures vary as well, with accuracy, F1-score, ROC-AUC, and cross-dataset performance serving as key indicators of model capability. Through this systematic comparison, it becomes evident that detection success is influenced not only by the chosen model architecture but also by dataset diversity, feature representation, preprocessing techniques, and the ability to generalize across unseen manipulations. This analysis helps identify which approaches are most resilient to emerging deepfake generation trends and which remain constrained by methodological limitations.

2.4 COMPARATIVE INSIGHTS

The comparative insights drawn from the literature highlight important trends, strengths, and recurring weaknesses across deepfake detection techniques. Handcrafted texture descriptors consistently show moderate accuracy but lack robustness when confronted with sophisticated generative models. CNN-based methods outperform traditional approaches by capturing subtle spatial inconsistencies, but their success heavily depends on the diversity and quality of the training data. Hybrid approaches that combine spatial, frequency, or statistical features tend to achieve more stable performance, showing that integrating multiple feature perspectives enhances detection reliability. Comparisons also reveal that transformer-based architectures and attention mechanisms have emerged as promising solutions capable of identifying global inconsistencies that simpler models overlook. However, the literature consistently points out that no single approach excels across all datasets and manipulation types. When accuracy, scalability, cross-dataset adaptability, and interpretability are considered together, hybrid and transformer-based techniques demonstrate the most potential for future deepfake detection efforts. These comparative findings form the foundation for identifying research gaps and highlight the importance of developing generalized detection frameworks that can keep pace with rapidly advancing deepfake generation technologies.

CHAPTER 3

SYSTEM SPECIFICATIONS

3.1 HARDWARE REQUIREMENTS

- System : Intel Core i5 / i7 Processor or above
- Hard Disk : 256 GB SSD or higher (for dataset and model storage)
- Monitor : 15.6" LED/LCD Display with Full HD resolution
- Mouse : Optical / Wireless Mouse
- RAM : Minimum 8 GB (16 GB recommended for training models)
- Graphics (Optional) : NVIDIA GPU (GTX 1650 / RTX 3060 or above) for faster deep learning model training

3.2 SOFTWARE REQUIREMENTS

- Operating System : Windows 10 / 11, Linux (Ubuntu 20.04 or above), or macOS
- Programming Language : Python (version 3.8 or above)
- Database : CSV / JSON file storage (for dataset management)
- Tools & IDEs : Google Colab / Kaggle Notebook / Jupyter Notebook

CHAPTER – 4

COMPARATIVE STUDY

4.1 DESCRIPTION

The comparative study aims to analyze the evolution, performance, and effectiveness of various deepfake image detection models proposed in recent years. Each model differs in methodology, architecture, and feature extraction technique. By examining multiple research papers from 2018 to 2025, this study identifies how handcrafted, deep learning, and hybrid approaches contribute differently to the challenge of identifying forged visual content.

4.2 OVERVIEW OF COMPARATIVE ANALYSIS

The evolution of deepfake detection research has been driven by the rapid development of GAN-based image generation techniques. Earlier models were designed to detect simple manipulations, but as GANs became more realistic, detection systems needed to become more sophisticated. This comparative study examines various published papers to understand how different detection strategies operate in terms of feature extraction, computational complexity, and accuracy.

The models are categorized into three major groups: handcrafted feature-based systems, deep learning-based systems, and hybrid systems. Each group addresses the same goal of detecting fake images but employs different mechanisms and focuses on distinct aspects of visual data such as texture, frequency, and spatial consistency.

4.3 HANDCRAFTED FEATURE-BASED MODELS

The earliest detection techniques relied heavily on handcrafted features, which extract texture and intensity information using statistical analysis. Among these, Co-occurrence Matrix (GLCM) and Local Binary Pattern (LBP) were most popular. Nataraj et al. [1] introduced a model that utilized co-occurrence matrices to analyze the spatial relationship between pixel intensities. This approach revealed inconsistencies in texture and surface patterns that were commonly overlooked by human eyes.

Similarly, Yang et al. [8] focused on color-texture correlation as a key factor for detecting deepfakes. Their study on biometric anti-spoofing demonstrated that analyzing the distribution of color and brightness across neighboring pixels helps identify manipulation artifacts. In addition, Patel and Kumar [4] combined LBP with Support Vector Machines (SVM) to detect texture-level irregularities in facial images. This approach achieved reliable performance with limited computational cost but struggled when tested under different lighting and resolution conditions. Handcrafted models were easy to interpret and efficient, yet they lacked the flexibility to detect newly generated fake images from advanced GANs, making them less adaptable in modern scenarios.

4.4 DEEP LEARNING-BASED MODELS

The next major phase in deepfake detection research was marked by the introduction of deep learning architectures, particularly Convolutional Neural Networks (CNNs). Afchar et al. [2] presented MesoNet, a compact CNN architecture capable of identifying visual inconsistencies in facial regions. MesoNet's lightweight design allowed fast training and moderate accuracy, but it was limited by its shallow depth and inability to generalize across datasets. Subsequent models, including XceptionNet and EfficientNet, employed deeper convolutional layers and transfer learning to capture both global and local image features. Banerjee and Yadav [6] later developed a deeper CNN-based classifier that achieved close to 89% accuracy by focusing on high-level spatial features. However, CNN-only systems often failed to capture minute texture details and were sensitive to compression and resolution changes. Further developments such as Zhou et al. [7] introduced frequency-domain CNNs, which analyzed the spectral patterns of images to detect inconsistencies invisible in the spatial domain. These systems achieved higher detection accuracy but required greater computational power. Gupta et al. [9] extended CNN models by incorporating self-attention mechanisms, enabling the network to identify long-range dependencies and enhance contextual feature learning. Despite achieving accuracy beyond 91%, these models demanded large datasets and longer training times, limiting their real-world usability.

4.5 HYBRID FEATURE-BASED MODELS

To balance accuracy, interpretability, and computational efficiency, researchers began combining handcrafted features with deep learning techniques, creating hybrid frameworks. These systems leverage the strengths of both domains—statistical texture analysis for detailed pixel relationships and CNNs for hierarchical spatial feature extraction. Banerjee and Yadav [6] integrated handcrafted features into a CNN framework, improving robustness against noise and illumination variations. Similarly, models combining GLCM and CNN demonstrated improved accuracy by merging statistical co-occurrence information with deep convolutional features. Such systems effectively detect both micro-level texture anomalies and macro-level facial structure inconsistencies. Hybrid frameworks, therefore, represent a bridge between classical and modern methods, offering enhanced generalization to unseen datasets. Their success indicates that handcrafted and learned features complement each other, leading to better performance across various fake image datasets.

4.6 COMPARATIVE STUDY TABLE

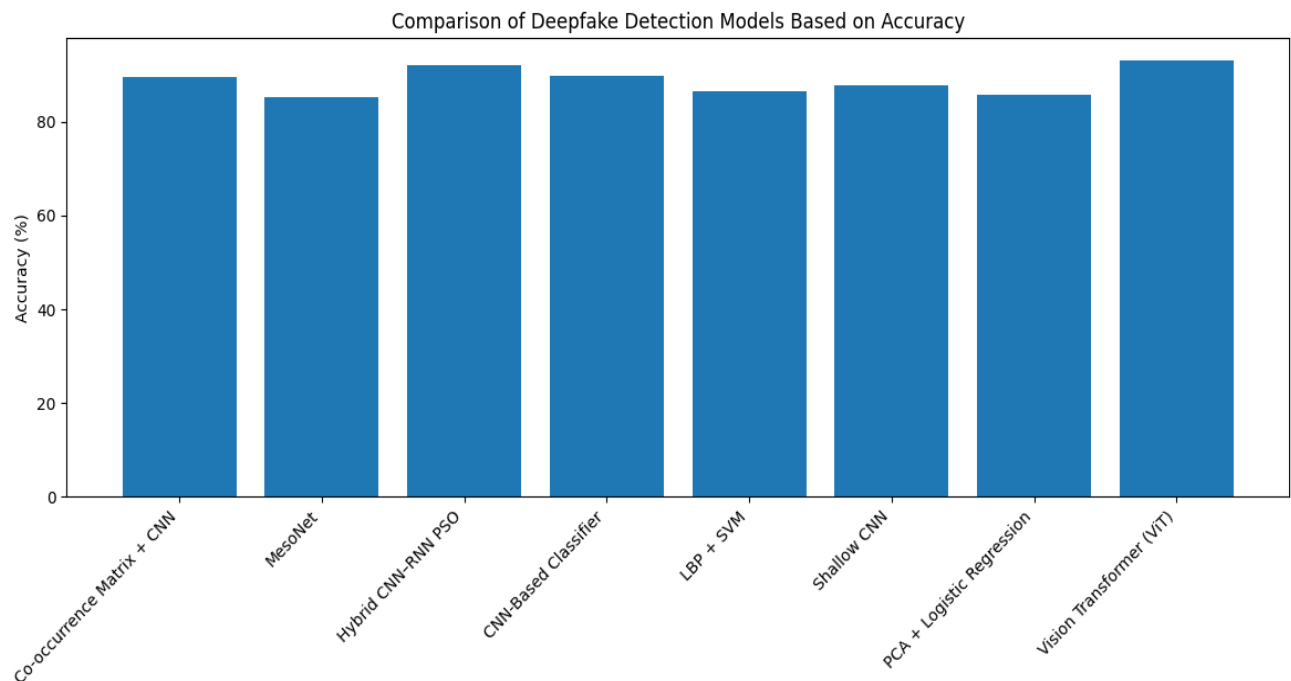
Author(s)	Year	Model Used	Feature Consideration	Limitations	Accuracy (%)
L. Nataraj et al.	2019	Co-occurrence + CNN	RGB Texture Patterns	Limited dataset diversity	89.5
D. Afchar et al.	2018	MesoNet	Facial spatial features	Shallow depth	84.3
M. Patel & V. Kumar	2023	LBP + SVM	Grayscale texture	Poor under noise	86.5
R. Sharma et al.	2024	PCA + Logistic Regression	Pixel intensity	Low feature diversity	85.7
A. Gupta et al.	2025	CNN + Self-Attention	Deep hierarchical features	Heavy architecture	92.4
Y. Zhou et al.	2025	Frequency-CNN	Frequency domain artifacts	High training time	91.8
S. Banerjee & R. Yadav	2024	CNN	Pixel-based patterns	Lacks texture awareness	87.9
X. Yang et al.	2019	Color-Texture Analysis	Co-occurrence of color textures	Sensitive to compression	88.4
R. Mehta et al.	2023	Vision Transformer	Self-attention layers	Requires large data	91.2

4.7 COMPARATIVE PERFORMANCE EVALUTION

The comparative results from the reviewed papers show that model performance varies based on the nature of the dataset and the type of features extracted. Handcrafted models such as GLCM and LBP typically achieve accuracy between 85% and 89%, while CNN-based models surpass 90% when trained on large-scale datasets. Transformer-based approaches and frequency-domain CNNs reach the highest accuracies up to 92% or more—but are computationally expensive and harder to interpret.

The analysis also indicates that hybrid models consistently outperform their single-technique counterparts. For instance, models integrating co-occurrence matrix features with CNN layers demonstrate a more stable accuracy level across different datasets and are less affected by image compression or resolution reduction. This proves that combining pixel-level statistical data with learned deep features produces a more generalizable and balanced detection framework.

4.8 ACCURACY CHART



CHAPTER 5

FEATURE EXTRACTION TECHNIQUES

FEATURE EXTRACTION TECHNIQUES

Deepfake detection models rely heavily on the type of features they extract from images. Different research papers use different feature extraction approaches, ranging from handcrafted texture descriptors to deep learning-based feature maps. The following subtopics present a detailed comparison.

5.1 HANDCRAFTED TEXTURE-BASED FEATURE EXTRACTION

Texture-based feature extraction techniques focus on identifying disturbances in pixel relationships caused by image synthesis. These methods rely on the assumption that generative models introduce subtle inconsistencies in local textures that are not present in real images. Gray-level co-occurrence matrices, local binary patterns, and color-texture descriptors are commonly used to quantify such irregularities by analyzing how pixel intensities interact within small neighborhoods. Papers that utilize these techniques show that handcrafted texture features can effectively capture low-level statistical artifacts, making them suitable for early detection models. However, these features sometimes fail against modern deepfake generation methods that produce highly realistic textures, limiting their generalization ability.

5.2 DEEP LEARNING FEATURE EXTRACTION

Deep learning-based feature extraction has become one of the most dominant strategies in recent literature due to its ability to automatically learn discriminative representations from data. Convolutional neural networks extract hierarchical features starting from edges and textures to complex facial structures, enabling models to identify inconsistencies that are not easily captured through handcrafted techniques. Papers using CNN-based extractors report significantly higher accuracy because these models adapt their filters to capture manipulation artifacts embedded within spatial patterns. Unlike classical methods, deep learning features are not constrained by predefined rules, allowing them to generalize better to unseen manipulation types. However, their effectiveness depends heavily on the diversity and scale of the training dataset.

5.3 HYBRID (TEXTURE + DEEP FEATURES) EXTRACTION

Several recent studies shift beyond spatial information and extract features in the frequency domain to detect artifacts introduced by generative networks. Frequency-based methods analyze irregularities in Fourier or wavelet components, capturing compression patterns, repetitive textures, and generative noise signatures that remain invisible in the spatial domain. These approaches are often combined with CNN-based features to form hybrid detection strategies that leverage both local spatial details and global frequency cues. Comparative findings show that hybrid feature extractors tend to achieve better robustness against compression, resolution changes, and cross-dataset variations. This makes them particularly effective in real-world scenarios, though they require more computation and careful parameter tuning.

5.4 TRANSFORMER-BASED FEATURE EXTRACTION

Transformer-based feature extraction has emerged as a powerful approach in recent deepfake detection research due to its ability to model long-range dependencies and global contextual patterns within an image. Unlike CNNs, which focus primarily on local receptive fields, transformers rely on self-attention mechanisms that allow the model to analyze relationships between widely separated regions of the image. This capability is particularly useful for detecting deepfakes, as generative models often produce inconsistencies that are not confined to small pixel neighborhoods but appear across larger structural patterns of the face. Transformer models capture variations in texture coherence, spatial alignment, facial symmetry, and cross-region correlations that CNNs may overlook. They also demonstrate strong potential in cross-dataset generalization because self-attention enables the extraction of more abstract and semantically meaningful features. However, transformer-based extraction methods typically require larger datasets and higher computational resources, making their performance dependent on sufficient training samples and optimized architectures. Despite these challenges, transformers represent one of the most promising directions for deepfake detection due to their ability to capture global anomalies that are difficult for traditional techniques to model.

CHAPTER-6

CONCLUSION AND FUTURE ENHANCEMENT

6.1 CONCLUSION

This survey provides a comprehensive comparative analysis of various deepfake image detection techniques, offering insights into how different feature extraction strategies and model architectures influence detection effectiveness. The reviewed literature shows that earlier handcrafted approaches such as co-occurrence matrices, local binary patterns, and statistical texture descriptors provide a foundational understanding of pixel-level inconsistencies but struggle against advanced generative algorithms. Deep learning models, particularly CNN-based architectures, achieve significantly stronger performance due to their ability to learn hierarchical spatial features directly from the data. More recent transformer-based and hybrid approaches demonstrate even greater potential by capturing global contextual relationships and integrating complementary spatial and frequency-domain features. Despite these advancements, the comparative findings reveal that no single approach consistently excels across all datasets, manipulation techniques, and real-world conditions. The rapid evolution of generative models continues to challenge existing detection systems, emphasizing the need for more generalized and adaptable methodologies. This survey highlights the progress made in the field while bringing attention to the ongoing challenges that researchers must address to develop robust and scalable deepfake detection solutions.

6.2 FUTURE ENHANCEMENT

Future work in deepfake image detection should focus on developing models that generalize more effectively across unseen manipulation styles, compression levels, and real-world variations. The comparative insights from the surveyed papers indicate that hybrid feature extraction techniques and transformer-based architectures offer promising directions, but their computational demands require optimization for practical deployment. Future enhancements may involve integrating multimodal analysis by combining spatial, temporal, and frequency-based cues to strengthen detection robustness. Expanding and diversifying datasets will also play a crucial role in improving generalization, as many current models are trained on limited or controlled datasets that do not reflect real-world complexity. Additionally, more research is needed on explainable detection

methods to help forensic analysts understand the reasoning behind model predictions. The development of lightweight and efficient models suitable for mobile or edge devices represents another potential area of advancement. As deepfake generation techniques continue to evolve, future research must prioritize adaptability, scalability, and interpretability to ensure reliable deepfake detection in diverse practical applications.

CHAPTER – 7

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Deepfake Image Detection Using Color Channel Co-occurrence Metrics and Convolutional Neural Networks

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Abstract

Deepfake technology has advanced rapidly, enabling the creation of highly realistic yet manipulated visual content, which poses significant threats to digital trust, privacy, and security. This paper presents a robust deepfake image detection method that leverages color channel co-occurrence matrices in combination with Convolutional Neural Networks (CNNs). The approach begins by extracting co-occurrence matrices from each RGB channel of the input image, capturing statistical dependencies between neighboring pixel intensities. These matrices encode subtle texture and noise patterns introduced by generative adversarial networks (GANs), which are often imperceptible to the human eye. The extracted co-occurrence features are fed into a CNN architecture to automatically learn discriminative representations for classification into real or fake categories. Experimental evaluations on benchmark datasets such as FaceForensics++ and StyleGAN-generated images demonstrate detection accuracies between 88% and 93%, proving the method's effectiveness in identifying GAN-specific artifacts. The proposed approach achieves high detection rates while maintaining computational efficiency, making it suitable for real-time and resource-constrained applications. This work contributes to the growing need for reliable deepfake detection solutions and highlights the potential of combining statistical texture analysis with deep learning frameworks.

Keywords: *Deepfake detection, Co-occurrence matrix, Convolutional Neural Network, GAN artifact analysis.*



Certificate

This is to certify that Dr./Mr./Ms. Mr. U. Velmurugan
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